

BE-2 — Start Here

Machine Learning. Layers 2–3. PS component D.

The full picture is in `work-breakdown.md`. This is what to do **today**, in order. Do not read the 29-page document first — do task 1, then read.

You are not blocked on BE-1, and you should not wait

Your models consume the **residual vector**, not raw telemetry. That vector is specified in `docs/spec/residual-spec.md` and you can generate synthetic residuals yourself in an afternoon. Start now.

More importantly, your **highest-value task of the entire week does not involve our data at all** — so do it first.

Task 1 — N-CMAPSS baseline, BEFORE touching our simulator (target: today)

This is the highest-value 90 minutes of your week and it is task one for a reason.

Get a RUL score on NASA's turbofan degradation data. Both datasets live in the NASA Prognostics Center of Excellence repository — "Turbofan Engine Degradation Simulation" is C-MAPSS, "...Simulation-2" is **N-CMAPSS**, which is the better one because it carries real flight conditions.

```
ml/benchmarks/ncmapss_baseline.ipynb
```

Why this comes first

It is **the only thing that proves our ML is not memorising our own simulator**. A DRDO panel will ask "*your data is synthetic, why should we believe any of it?*" and the answer ends with "*and here it is running unmodified on NASA's flight-condition data, with the score.*" That sentence is worth building for.

It also de-risks your whole week. If everything else goes wrong, you still have the single most credible slide in the deck.

Done means: a number on a slide, and the notebook committed.

Report the field's own metrics, not just RMSE

RMSE alone reads like a course assignment. Add:

- **NASA asymmetric score** — $\sum e^{(-d/13)} - 1$ if $d < 0$, else $\sum e^{(d/10)} - 1$. Late predictions punished harder than early. Field standard.
 - **Prognostic horizon** and **α - λ accuracy** — standard PHM measures; using them signals you know the discipline.
 - **False alarms per flight hour** — *the question a maintenance officer asks first*. Quoting it unprompted is a strong signal.
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Task 2 — The rule that governs everything you build

Every model you build eats the residual vector ρ . Never raw sensor values.

This is not a stylistic preference and you will be asked to defend it.

An autoencoder trained on raw signals learns the **flight profile**, not the engine. Then every throttle movement and every climb changes every channel, and it screams. **That is the single biggest reason naive builds false-alarm constantly**, and a lot of teams will discover it on the last day.

The residual vector is **nominally zero** and **operating-point invariant by construction**, so your autoencoder learns a genuinely simple target — *healthy means near zero and white* — and stays quiet through a climb.

You should be able to say that paragraph from memory. It answers at least two judge questions.

And prove it: same autoencoder trained on raw signals vs on residuals, one bar chart. That is the **ablation** (day 4), and it converts our central design claim from an assertion into a measured result. One afternoon.

Task 3 — Three models, each with one job (days 2–3)

Resist building five model families because the problem statement mentions several.

M1 — physics-guided residual estimator

$$\hat{y} = \underbrace{f_{\text{physics}}(u; \theta)}_{\text{MVEM}} + \underbrace{g_{\text{NN}}(u)}_{\text{small correction}}$$

The network learns only the **discrepancy**, so it trains on very little data and **cannot produce absurd predictions at operating points outside its training set** — which matters, because your training data will be mid-altitude cruise and the aircraft flies to 28 000 ft.

Soft physical constraints in the loss:

$$L = L_{\text{data}} + \lambda_1 \|p_4\|^2 + \lambda_2 \|\dot{m}_a + \dot{m}_f - \dot{m}_{\text{ex}}\|^2 + \lambda_3 \|\min(0, \partial T_{\text{cht}} / \partial \dot{m}_f)\|$$

energy mass continuity monotonicity

On PINNs, honestly: do the *soft* version. A strict PDE-residual PINN is a research project that may not converge in our window, and if it doesn't we have nothing. What we claim — physics backbone, learned correction, physically motivated loss terms — is what we deliver, and the docs say so plainly.

Virtual sensors this gives you, and DRDO asks for virtual sensing by name: peak cylinder pressure, knock margin, per-cylinder combustion efficiency, turbo shaft speed, oil film thickness.

M2 — LSTM autoencoder on residuals

Train on healthy windows only. Threshold is the **99.5th percentile of reconstruction error on held-out healthy data** — *never* a hand-picked constant, and be ready to say that out loud. Add an N-of-M persistence rule to suppress transients.

M3 — classifier + RUL

1D-CNN or GRU over the residual window → softmax across the ten fault classes. Labels are free because BE-1 generates the faults.

Run it alongside the incidence-matrix match and show both agreeing. The network supplies accuracy; the matrix supplies the explanation. *Agreement between two independent mechanisms is itself evidence*, and judges remember the why.

RUL — two heads, always both: a quantile network (p10/p50/p90) and a physics head integrating damage forward. Report both, advise on the conservative. *"Physics says 4.1 hours, the network says 3.7, we advise on 3.7."* Disagreement beyond the interval is surfaced as a warning in its own right.

Task 4 — Novelty detection (day 3) — NEW, and the strongest thing you own

Not in the original scope. About **forty lines**. It answers the most dangerous question a panel can ask:

"You have ten faults in your library. What happens when the engine does something that isn't one of them?"

Every classifier is forced to pick from its list, so the usual answer is that it confidently names the closest wrong thing.

Each fault signature is a **direction** in the 11-dimensional residual space. Stack them into **F** and split the live residual:

```
 $\hat{\rho} = \text{proj\_span}(F) \rho$       explained  
 $\rho_{\perp} = \rho - \hat{\rho}$           UNEXPLAINED  
 $v = \|\rho_{\perp}\| / \|\rho\|$ 
```

High $\|\rho\|$ with high v means something real that the library **cannot express**. The honest output is *low confidence*, not a confident wrong answer.

Reference implementation to port: `frontend/src/analysis/novelty.ts`. It works and it is running in the dashboard now.

Four things that will bite you, all already found

1. **Use the SVD.** Not normal equations — **F** is ill-conditioned by construction.
2. **rho6..rho9 is NOT one direction.** The incidence matrix lists them as a group, which is right for a printed table and wrong as a direction in residual space. A single-cylinder fault lifts **its own** thermal deviation and pushes the other three slightly negative, because each is measured against the **conditional mean**. Sum-to-zero is baked into the residual definition.

Treated as a block, a genuine in-library fault does not lie in $\text{span}(F)$ at all. Measured:

	nu BEFORE	nu AFTER
EGT sensor drift, cyl 4 (in library)	0.932	0.000
Injector fouling, cyl 2 (in library)	0.610	0.000
Unmodelled fault (null space)	0.000	1.000

It was crying wolf on two thirds of its own library while scoring the genuinely unknown fault at zero — exactly backwards. Expand per-cylinder faults into one column each, thermal part shaped as $(e_i - \text{mean})$.

3. **Weight v by residual significance.** Unweighted it reads ~50% confidence on a *healthy* engine, because isotropic noise puts $\sqrt{(3/11)}$ of itself in the null space by construction. A confidence channel that panics at idle gets ignored when it matters.

4. **RECOMPUTE THE RANK ON THE SENSITIVITY MATRIX.** Everything above comes from the coarse glyph matrix (entries 0, ± 1 , ± 2). The physically correct object is the **Jacobian of each residual with respect to each fault parameter**, with continuous entries. On the coarse matrix, correcting item 2 already took the null space from 3 dimensions to **1**. **There is a real chance it goes to 0 on the continuous matrix — in which case we drop the claim and say why.** Check before anyone builds a slide on it.

Full spec: `docs/spec/novelty-detection.md`.

Definition of done, per day

DAY	DONE MEANS
1	N-CMAPSS baseline score. Notebook committed.
2	M1 + M2 trained on synthetic residuals, threshold calibrated on healthy held-out data.
3	M3 classifier + confusion matrix. Two-headed RUL with p10/p50/p90. Novelty projection ported with the SVD rank report .
4	Measurement day: false-alarm rate over 100 healthy flights, detection lead time, the raw-vs-residual ablation.
5	Edge latency benchmark. Pre-trained weights committed to the repo.

What you must not do

- **Do not** train on raw sensor values. Ever. Residuals only.
 - **Do not** hand-pick an anomaly threshold. Calibrate on healthy held-out data and be able to say how.
 - **Do not** report a bare RUL number without its uncertainty band.
 - **Do not** attempt a strict PDE-residual PINN. Scoped out deliberately, and we say why.
 - **Do not** train during the demo. Weights are committed.
 - **Do not** put the rank-10 figure on a slide until you have recomputed it on the sensitivity matrix.
 - **Do not build the GNN** unless day 4 closed completely clean — and then only as a second isolation head *beside* the signature matrix, never replacing it.
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Where to ask

- The residual vector you consume → `docs/spec/residual-spec.md`
- Novelty maths → `docs/spec/novelty-detection.md`
- Why any of this → `docs/SOLUTION-ANALYSIS.md` §2.9–2.11